

The diffuse domain method on a mass-conservative adaptive multigrid solver: formulation, well-posedness, and ε -convergence analysis

DiffuseDomainBSAM documentation
companion to `EllipticBSAM/docs/convergence_theory.pdf`

Abstract

We analyze the diffuse-domain reformulations solved by `DiffuseDomainBSAM`: the flux/Neumann form of Feng, Guo, Lowengrub & Wise (2018), eq. (6.18), and a Dirichlet penalty form, both of which are instances of the general solver's equation $-\nabla \cdot (D_{dd} \nabla u) + C_{dd} u = f_{dd}$ with particular coefficients built from a phase field ϕ . We prove uniform ellipticity of the floored formulation and explain the formal matched-asymptotic prediction of $O(\varepsilon^2)$ for the smooth Neumann model under the required extension and matching assumptions. This is not a uniform error theorem for every coefficient, geometry, or penalty formulation. We also prove an exact-reproduction lemma: for the analytic data extensions used in the classical star-domain benchmark, the continuous diffuse-domain solution *satisfies the unfloored differential equation* for every ε . Equality of boundary-value solutions additionally requires compatible outer data; the implemented floor is a separate perturbation. A constant-normal extension gives measured rates near two in a small circle study. We further quantify how the adaptive hierarchy with geometric band tagging resolves the layer at the finest resolution while leaving the bulk coarse, and why the flux-balance corrections of the underlying solver are essential at small ε .

1 Formulation

Let $\Omega_1 \subset \mathcal{B}$ be a smooth domain compactly contained in the rectangle $\mathcal{B}$, described by a level set ψ with $\psi < 0$ in Ω_1 , $\Gamma = \{\psi = 0\}$. Define the phase field

$$\phi(x) = \frac{1}{2} \left(1 - \tanh \frac{3\psi(x)}{\varepsilon} \right), \quad (1)$$

so $\phi \rightarrow 1$ in Ω_1 , $\phi \rightarrow 0$ outside, with transition layer of width $O(\varepsilon)$. Two reformulations of

$$\beta u - \nabla \cdot (D \nabla u) = f \text{ in } \Omega_1, \quad (\text{BC on } \Gamma) \quad (2)$$

onto $\mathcal{B}$ (homogeneous Neumann on $\partial\mathcal{B}$) are used:

Neumann data $D \partial_n u = g$ on Γ (Feng et al. eq. (6.18); Li–Lowengrub–Rätz–Voigt [2]):

$$\beta \phi u_\varepsilon - \nabla \cdot (\phi D \nabla u_\varepsilon) = \phi f + |\nabla \phi| g \quad \text{in } \mathcal{B}, \quad (3)$$

i.e. the general solver is called with $D_{dd} = \phi D$, $C_{dd} = \beta \phi$, $f_{dd} = \phi f + |\nabla \phi| g$.

Dirichlet data $u = g$ on Γ (penalty form):

$$-\nabla \cdot (\phi D \nabla u_\varepsilon) + \phi C u_\varepsilon + \frac{\mu}{\varepsilon^3} (1 - \phi) (u_\varepsilon - g) = \phi f, \quad (4)$$

i.e. $C_{dd} = \phi C + \mu \varepsilon^{-3} (1 - \phi)$, $f_{dd} = \phi f + \mu \varepsilon^{-3} (1 - \phi) g$.

Here f and g denote smooth extensions of the data off Γ .

2 Well-posedness and the ϕ -floor

In double precision tanh saturates and ϕ vanishes identically a few layer-widths outside Γ , making (3) degenerate. The implementation replaces ϕ by $\phi_\delta = \max(\phi, \delta)$ in D_{dd} (and in C_{dd} for (3)), $\delta = 10^{-8}$ by default.

Lemma 2.1 (continuous coercivity at fixed floor). *Suppose $0 < D_{\min} \leq D \leq D_{\max}$ and $0 < \delta < 1$. Then $D_{dd} \geq \delta D_{\min} > 0$. For the Neumann formulation with $\beta > 0$, $C_{dd} \geq \beta \delta > 0$, and its weak bilinear form is coercive on $H^1(\mathcal{B})$, hence has a unique solution for bounded data. For the penalty formulation assume $C \geq 0$, $\mu > 0$, and a positive-measure exterior region; its positive reaction controls the constant mode and a weighted Poincaré inequality gives the same fixed-parameter conclusion. If $\beta = 0$ with only Neumann outer data, compatibility and a mean normalization are required. These continuous statements do not establish that an assembled QIF composite matrix is an M -matrix: interpolation, coefficient ratios, and mesh structure must also be checked.*

Lemma 2.2 (effect of the floor). *Let $0 < \delta < 1/2$. Let u_ε solve (3), and let u_ε^δ solve the equation with diffusion $\phi_\delta D$ and reaction $\beta \phi_\delta$, keeping the assembled right-hand side $f_{dd} = \phi f + |\nabla \phi| g$ unchanged, as in the implementation. The coefficient perturbation is supported in $\{\phi < \delta\} = \{\psi > \psi_\delta\}$ with $\psi_\delta = \frac{\varepsilon}{6} \ln \frac{1-\delta}{\delta} \approx \frac{\varepsilon}{6} \ln \frac{1}{\delta}$, a logarithmic distance in layer units outside Γ . Assume $\beta > 0$, homogeneous outer Neumann data, and the stated ellipticity bounds on D . Testing the difference equation with the difference and using $\phi \leq \phi_\delta \leq \phi + \delta$,*

$$\|\sqrt{\phi_\delta} \nabla (u_\varepsilon^\delta - u_\varepsilon)\|_{L^2(\mathcal{B})}^2 + \beta \|\sqrt{\phi_\delta} (u_\varepsilon^\delta - u_\varepsilon)\|_{L^2(\mathcal{B})}^2 \leq C \delta (\|\nabla u_\varepsilon\|_{L^2(\phi < \delta)}^2 + \|u_\varepsilon\|_{L^2(\phi < \delta)}^2),$$

Since $\phi \geq 1/2$ inside Ω_1 , this controls the interior H^1 error by $O(\delta^{1/2})$ times the displayed exterior norms. Uniformity in ε requires an independent bound on those norms. The estimate is an energy estimate, not an L^∞ bound, and it does not certify that a particular floor is below a requested tolerance.

3 Sharp-interface limit: matched asymptotics

We expand (3). Assume first that $\psi = d$, the signed distance to Γ ($|\nabla d| = 1$), let κ be the curvature, and use normal-tangential coordinates (d, s) with the stretched variable $z = d/\varepsilon$ and profile $\Phi(z) = \frac{1}{2}(1 - \tanh 3z)$, so that

$$\phi = \Phi(z), \quad |\nabla \phi| = \varepsilon^{-1} |\Phi'(z)|, \quad \int_{-\infty}^{\infty} |\Phi'(z)| dz = 1.$$

Write outer expansions $u_\varepsilon = u^0 + \varepsilon u^1 + \dots$ in Ω_1 (and in $\mathcal{B} \setminus \overline{\Omega}_1$) and an inner expansion $\tilde{u}(z, s) = \tilde{u}^0 + \varepsilon \tilde{u}^1 + \varepsilon^2 \tilde{u}^2 + \dots$. In the layer,

$$\nabla \cdot (\phi D \nabla u) = \varepsilon^{-2} \partial_z (\Phi D \partial_z \tilde{u}) + \varepsilon^{-1} \kappa \Phi D \partial_z \tilde{u} + O(1).$$

Order ε^{-2} . $\partial_z(\Phi D \partial_z \tilde{u}^0) = 0$, so $\Phi D \partial_z \tilde{u}^0 = a(s)$. Matching to the interior outer solution as $z \rightarrow -\infty$ ($\Phi \rightarrow 1$, $\partial_z \tilde{u}^0 \rightarrow 0$) forces $a \equiv 0$; since $\Phi > 0$ for all finite z , $\partial_z \tilde{u}^0 \equiv 0$:

$$\tilde{u}^0 = \tilde{u}^0(s) = u^0|_{\Gamma}(s), \quad (5)$$

the leading inner solution is constant across the layer. The interior outer problem at $O(1)$ is the sharp equation (2).

Order ε^{-1} — recovery of the Neumann condition. Using (5),

$$-\partial_z(\Phi D \partial_z \tilde{u}^1) = |\Phi'(z)| g(s).$$

Integrating over $z \in \mathbb{R}$ and using $\Phi D \partial_z \tilde{u}^1 \rightarrow D \partial_n u^0|_{\Gamma}$ as $z \rightarrow -\infty$ (matching) and $\rightarrow 0$ as $z \rightarrow +\infty$ (boundedness; verified below), and $\int |\Phi'| = 1$:

$$D \partial_n u^0 = g \quad \text{on } \Gamma. \quad (6)$$

Thus u^0 solves the sharp problem. Solving for the profile,

$$\Phi D \partial_z \tilde{u}^1(z) = D \partial_n u^0|_{\Gamma} - g \int_{-\infty}^z |\Phi'| = g \int_z^{\infty} |\Phi'| dz',$$

whose right side decays like e^{-6z} , matching $\Phi \sim e^{-6z}$, so $\partial_z \tilde{u}^1$ is bounded: the expansion is consistent.

Next order and scope of the formal prediction. For the smooth, constant-coefficient Neumann model with signed distance and constant-normal extensions, the full matching calculation in [3] gives $u^1 = 0$ when $\beta > 0$; in the pure Neumann case the constant correction must be fixed by a consistent gauge. The symmetric profile satisfies $\int_{\mathbb{R}} z |\Phi'(z)| dz = 0$ and $\int_{\mathbb{R}} (\Phi(z) - \mathbf{1}_{z < 0}) dz = 0$. These identities help cancel the first correction; by themselves they are not a substitute for all solvability and matching conditions. The resulting formal prediction is

$$\|u_{\varepsilon} - u\|_{L^2(\Omega_1)} = O(\varepsilon^2), \quad (7)$$

under those assumptions. Extending it to variable D , an arbitrary data extension, a finite outer boundary, the floor, and a changing adaptive mesh requires additional error estimates.

Lemma 3.1 (exact reproduction by compatible extensions). *Let $u_e \in C^2(\bar{\mathcal{B}})$ and suppose the data extensions satisfy, pointwise in $\mathcal{B}$,*

$$\beta u_e - \nabla \cdot (D \nabla u_e) = f \quad \text{and} \quad |\nabla \phi| g = -\nabla \phi \cdot D \nabla u_e.$$

Then u_e satisfies the unfloored differential equation (3) exactly, for every ε : indeed $\beta \phi u_e - \nabla \cdot (\phi D \nabla u_e) = \phi(\beta u_e - \nabla \cdot (D \nabla u_e)) - \nabla \phi \cdot D \nabla u_e = \phi f + |\nabla \phi| g$. The second condition holds whenever g is extended as $g = D \nabla u_e \cdot \nabla \psi / |\nabla \psi|$, because $\nabla \phi = \phi'(3\psi/\varepsilon)^{\frac{3}{\varepsilon}} \nabla \psi$ with $\phi' < 0$, so $|\nabla \phi| = -\phi'^{\frac{3}{\varepsilon}} |\nabla \psi|$. In particular, for the test data $u_e = (x^2 + y^2)/4$, $f = u_e - \Delta u_e = r^2/4 - 1$ and $g = \nabla u_e \cdot n_{\psi}$ (any geometry, including the flower), this is an exact differential identity. To be the solution on $\mathcal{B}$, u_e must also satisfy the imposed outer boundary condition. The quadratic does not satisfy homogeneous Neumann conditions on a finite box, and the floor changes the identity. Thus the benchmark is dominated by layer discretization when these other errors are

small, but does not have identically zero total model error. This observation also applies to the corresponding test construction of Feng et al. (their Sec. 6.4): the ε -convergence reported there is convergence of the numerical solution under the coupled (ε, h) schedule. To probe the genuine $O(\varepsilon^2)$ model convergence one must use an extension that breaks the identity — the canonical choice being the constant-in-normal extension $g \equiv g|_\Gamma$ — and separate the discretization error, e.g. by Richardson extrapolation in h at fixed ε (test D1b).

Remark 3.2 (non-distance level sets). If Γ is described by a level set ψ with $|\nabla\psi| \neq 1$ (e.g. the flower $\psi = r - R_0(\theta)$ of Feng et al. eq. (6.19)), then in the true normal coordinate $\psi = |\nabla\psi|_\Gamma d + \frac{1}{2}(\partial_n^2\psi) d^2 + O(d^3)$, and $\phi = \Phi(\psi/\varepsilon)$ has a spatially varying physical width and can be skewed in d . The signed-distance calculation above therefore cannot be transferred without re-expanding the geometry and data. No universal rate between one and two follows from $|\nabla\psi| \neq 1$ alone: even a constant rescaling of a distance function changes the width without destroying symmetry. Reinitialization can simplify the analysis, but does not itself prove a convergence rate.

Remark 3.3 (corners). For a polygonal Ω_1 with the exact signed distance as level set, $|\nabla\psi| = 1$ a.e. but $\nabla\psi$ jumps across the corner bisectors. Two distinct effects follow. (i) In the *flux* form (3) the extended datum $g = \nabla u \cdot n_\psi$ inherits the jump: f_{dd} is discontinuous along five lines (for a pentagon) inside the band, and the corner neighborhoods converge at reduced order in L^2 (measured: rate ≈ 0.7 in L^2 , ≈ 1.7 in L^∞ , at fixed ε/h) in the recorded test. Discontinuous extended data can reduce regularity; these observations alone do not isolate the source of the measured rate. (ii) The *penalty* form (4) never references the normal; corners enter only through the geometry of ϕ , and the measured ε -rates on the pentagon are 1.5–1.8 over the recorded range. The smooth Neumann expansion does not establish the asymptotic penalty rate at corners.

Dirichlet penalty. For (4) the dominant inner balance at $O(\varepsilon^{-3})$ is $\mu(1 - \Phi)(\tilde{u}^0 - g) = 0$; since $1 - \Phi > 0$ for all finite z , $\tilde{u}^0 \equiv g$ throughout the layer and matching gives $u^0 = g$ on Γ : the Dirichlet condition is enforced at leading order. The ε^{-3} scaling makes the penalty dominate the diffusion (ε^{-2}) by one order, and the leading-order argument does not determine the error rate. The paper [3] analyzes Neumann and Robin approximations and is not a proof of second-order accuracy for this Dirichlet penalty. The disc suite measures rates 1.8, 2.1, and 3.6 over its finite schedule; these are observations for that problem, not a general second-order theorem.

4 Interaction with the adaptive solver

Remark 4.1 (band tagging and actual coverage). Tag every cell with $|\psi| < w\varepsilon$ (TagWidth $w = 2.5$ by default) on every level, with buffer $b \geq 2$ cells. Then outside the tagged band $|\phi - \mathbf{1}_{\psi < 0}| \leq \frac{1}{2}(1 - \tanh 3w) \approx 3 \cdot 10^{-7}$ ($w = 2.5$). This is a phase-field tail bound, not a guarantee of finest-level coverage: tags are sampled at cell centers, clustering may drop thin fragments, and the level cap may stop refinement. Check the visible cells intersecting the band. For narrow layers, a useful level-dependent tag width is $\max(w\varepsilon, 3h_k)$ for signed distance; this reduces missed coarse-grid fragments, but still needs coverage verification. The tail is nonzero even outside the tagged band.

Remark 4.2 (a local coefficient-ratio check). For two sample points separated by at most h , assume $|\nabla\psi| \leq G$ along the connecting segment. Since $|\partial_\psi \log \phi| \leq 6/\varepsilon$, the phase-field ratio

is bounded below by $e^{-6Gh/\varepsilon}$; flooring preserves this bound. If the physical diffusivity ratio is bounded below by $\rho_D > 0$, a sufficient check for a local ratio $D_2/D_1 \geq 1/5$ is

$$\rho_D e^{-6Gh/\varepsilon} \geq \frac{1}{5}.$$

For constant physical diffusivity and signed distance this reduces to $\varepsilon/h \geq 6/\log 5 \approx 3.73$. Use the actual spacing and geometry at the coarse–fine stencil, not just the finest spacing. This local check does not certify diagonal dominance or positivity of the entire composite QIF matrix.

Remark 4.3 (why the flux-balance corrections matter here). The DDM coefficients are exponentially small outside the band, so the solution is determined almost entirely through the layer — the region crossed by every coarse–fine interface. Any flux imbalance there acts like a spurious source of relative size $O(h^2)$ *concentrated exactly where all the physics happens*. The suite’s ablation shows the consequence at $\varepsilon = 0.1$ (4-level mesh): disabling the corrections inflates the interior L^2 error from $3.0 \cdot 10^{-6}$ to $5.2 \cdot 10^{-3}$ (a factor ~ 1700) and the mass defect from 10^{-13} to $2 \cdot 10^{-3}$ — the multilevel analogue of the paper’s Table 6, and the mechanism behind the divergence of earlier in-house attempts at ≥ 3 levels.

5 Numerical verification

Produced by `tests/runDdTests.m`; complete logs and tables in `TEST.REPORT.md`. Highlights (interior L^2 errors against the sharp solution):

- **Flower (paper Sec. 6.4 replication), Neumann form, analytic extensions:** Lemma 3.1 explains the differential cancellation, subject to its floor and outer-boundary qualifications. The (ε, h) -schedule errors are dominated by layer discretization, decaying with observed rates ≈ 1.5 in ε at fixed $\varepsilon/h = 20.5$ down to $1.1 \cdot 10^{-6}$ at $\varepsilon = 0.05$ (effective 2048^2); the adaptive 4-level solution agrees with the uniform solution of equal finest resolution to $\lesssim 1\%$ while using 5–10 $\times$ fewer cells.
- **Circle, constant-normal extension (model-error diagnostic):** The diagnostic $(4E(h/2) - E(h))/3$ assumes a smooth h^2 expansion of the error norm; it estimates a model-error floor but is not a certified extrapolation of the solution. Its measured ε -rate is near two, consistent with (7). The displayed ratios of total errors at fixed ε are near zero order because model error dominates; they do not measure spatial order.
- **Disc, Dirichlet penalty:** rates 1.8–2.1 (with a superconvergent last interval as the discretization error crosses the model error).
- **Conservation:** relative mass defect 10^{-16} – 10^{-13} at all ε with the corrections on; up to 10^{-3} with them off, accompanied by a $\sim 10^3 \times$ loss of interior accuracy at $\varepsilon \leq 0.1$ on the 4-level meshes.
- **Solver:** V(3,3) factors 0.05–0.16 per cycle on the adaptive DDM hierarchies, independent of ε despite coefficient degeneracy of eight orders of magnitude.

References

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- [2] X. Li, J. Lowengrub, A. Rätz, A. Voigt, Solving PDEs in complex geometries: a diffuse domain approach, *Commun. Math. Sci.* 7 (2009) 81–107.
- [3] K. Y. Lervåg, J. Lowengrub, Analysis of the diffuse-domain method for solving PDEs in complex geometries, *Commun. Math. Sci.* 13 (2015) 1473–1500.